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5130904/30002

Вар 47

Среднеквадратичная аппроксимация

$$f(x) = \sin(\pi x), x \in [-\frac{1}{4}, \frac{1}{4}], n=2; Q_2 = d_0 \varphi_0(x) + d_1 \varphi_1(x) + d_2 \varphi_2(x)$$

$$\varphi_0(x) = 1; \varphi_1(x) = x; \varphi_2(x) = x^2$$

$$(\varphi_0, \varphi_0) = \int_{-\frac{1}{4}}^{\frac{1}{4}} dx = x \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}; (f, \varphi_0) = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x) dx = -\frac{\cos(\pi x)}{\pi} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = 0$$

$$(\varphi_0, \varphi_1) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x dx = \frac{x^2}{2} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{2} \left(\frac{1}{16} - \frac{1}{16} \right) = 0; (f, \varphi_1) = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x) x dx = -\frac{\sin(\pi x)}{\pi^2} - \frac{x \cos(\pi x)}{\pi} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = 0$$

$$(\varphi_0, \varphi_2) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 dx = \frac{x^3}{3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{3} \left(\frac{1}{64} + \frac{1}{64} \right) = \frac{1}{96} \quad \Leftrightarrow -\frac{\sqrt{2}}{4\pi} + \frac{\sqrt{2}}{\pi^2} \approx 0,03075$$

$$(\varphi_1, \varphi_0) = (\varphi_0, \varphi_1) = 0$$

$$(f, \varphi_2) = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x) x^2 dx = \frac{2x \sin(\pi x)}{\pi^2} - \frac{x^2 \cos(\pi x)}{\pi} + \frac{2 \cos(\pi x)}{\pi^3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = 0$$

$$(\varphi_1, \varphi_1) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x \cdot x dx = \frac{x^3}{3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{96} \quad \Leftrightarrow 0$$

$$(\varphi_1, \varphi_2) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 dx = \frac{x^3}{3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{256} \left(\frac{1}{16} - \frac{1}{16} \right) = 0$$

$$(\varphi_2, \varphi_0) = (\varphi_0, \varphi_2) = \frac{1}{96}$$

$$(\varphi_2, \varphi_1) = (\varphi_1, \varphi_2) = 0$$

$$(\varphi_2, \varphi_2) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 x^2 dx = \frac{x^5}{5} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{1}{5} \left(\frac{1}{1024} + \frac{1}{1024} \right) = \frac{1}{2560}$$

$$\begin{cases} (\varphi_0, \varphi_0)d_0 + (\varphi_0, \varphi_1)d_1 + (\varphi_0, \varphi_2)d_2 = (f, \varphi_0) \\ (\varphi_1, \varphi_0)d_0 + (\varphi_1, \varphi_1)d_1 + (\varphi_1, \varphi_2)d_2 = (f, \varphi_1) \\ (\varphi_2, \varphi_0)d_0 + (\varphi_2, \varphi_1)d_1 + (\varphi_2, \varphi_2)d_2 = (f, \varphi_2) \end{cases} \Leftrightarrow \begin{cases} \frac{1}{2}d_0 + \frac{1}{96}d_2 = 0 \\ \frac{1}{96}d_1 = -\frac{\sqrt{2}}{4\pi} + \frac{\sqrt{2}}{\pi^2} \\ \frac{1}{96}d_0 + \frac{1}{2560}d_2 = 0 \end{cases} \Leftrightarrow \begin{cases} d_0 = 0 \\ d_1 = \frac{96\sqrt{2}}{\pi^2} - \frac{24\sqrt{2}}{\pi} \\ d_2 = 0 \end{cases}$$

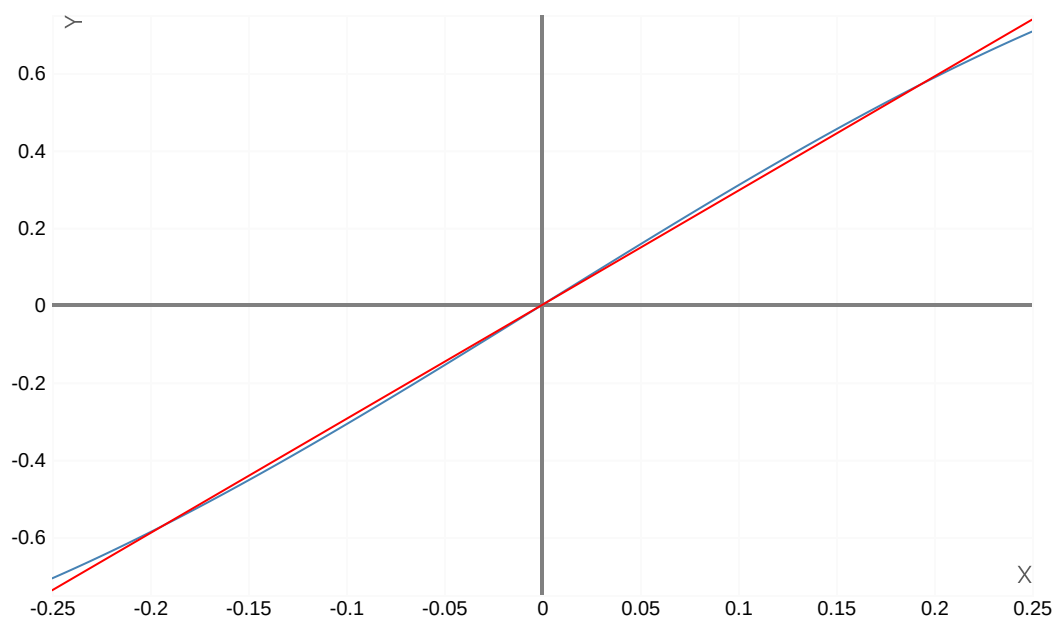
$$Q_2(x) = -\frac{24\sqrt{2}}{\pi}x + \frac{96\sqrt{2}}{\pi^2}x \approx 2,952x$$

$$E(x) = f(x) - Q_2(x) = \sin(\pi x) - 2,952x$$

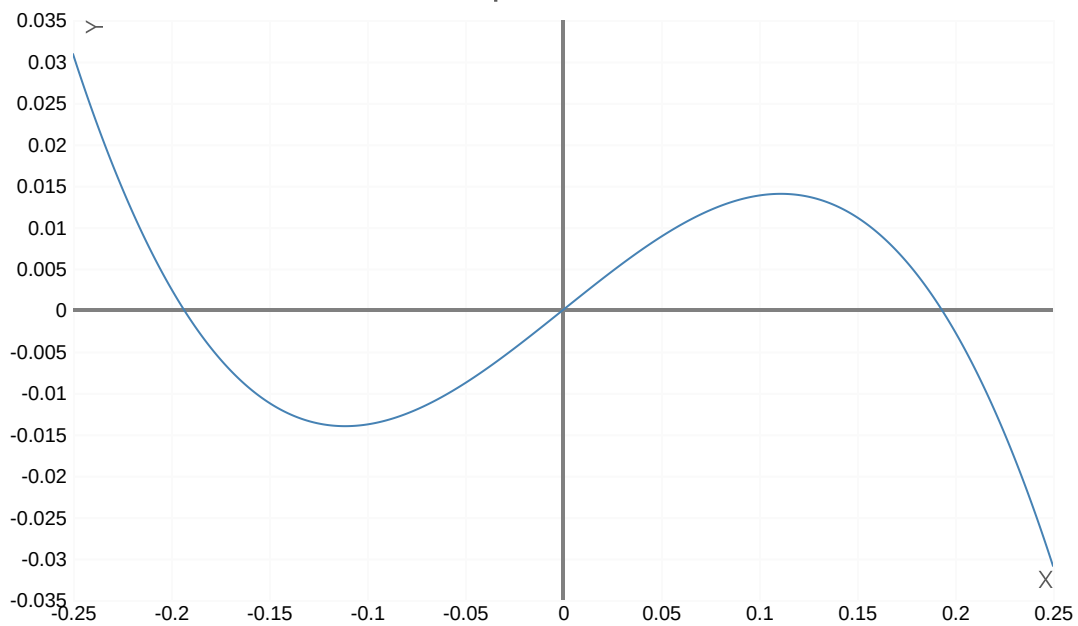
$$E^2(x) = \sin^2(\pi x) - \frac{738x \sin(\pi x)}{125} + 8,714x^2$$

$$\rho = \frac{1}{b-a} \int_a^b E^2(x) dx = \frac{1}{\frac{1}{4} + \frac{1}{4}} \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin^2(\pi x) dx = \frac{738x \sin(\pi x)}{125} + 8,714x^2 \approx 1,3 \cdot 10^{-4}$$

$\sin(\pi \cdot x); \mathbb{Q}[2]x$



$\epsilon(x)$



$$\epsilon(x)^2$$

